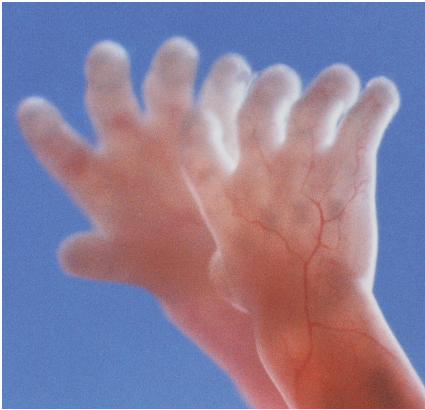




YOU ARE **22 WEEKS AND 2 DAYS**

124 days to go

YOUR BABY TODAY

The hands and fingers are clearly visible and the nail beds have been laid down. The baby's fingers will close if the palm of the hand is touched. The tips of the two bones in the forearm, the radius and the ulna, can just be seen in the lower part of this image.

“Now that your baby's ears are sufficiently well developed to process sounds, their hearing will gradually improve.”

Your baby's external ears have been developed for some time but for them to hear, the internal ear structures also need to mature. In the middle ear, three bones—the malleus (“hammer”), incus (“anvil”), and stapes (“stirrup”)—conduct sound into the inner ear. These bones are formed initially from soft cartilage and embedded within connective tissue. The bones begin to harden and the connective tissue gradually dissolves. This allows the ear drum to vibrate onto the hammer, which passes the movement on to the anvil, and then the stirrup. The vibrations are then passed to the cochlea, a cavity of the inner ear, where they are translated into nerve impulses to be sent to the brain.

At 22 weeks, your baby's inner ear has matured adequately for sound to be processed into neural signals to the brain. The first part of the cochlea to develop is responsible for receiving lower-sound frequencies. As your baby develops, they will gradually be able to recognize and respond to higher sound frequencies. Over the next three weeks, your baby's responsiveness to sounds will gradually increase. At first the responses are slow and sluggish, but by 25 weeks, they will react to a range of sounds by moving around. In addition to being responsible for your baby's hearing, the inner ear also controls their balance. Small fibers within three semicircular canals of the inner ear are able to sense acceleration in any direction, providing the sense of motion and balance. Floating in the amniotic fluid is similar to weightlessness and, although your baby is very active, they have no sense yet of moving up and down.